EZZ ELDIN EMAD

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EDUCATION

B.Sc. Computing & Data Science | Alexandria University - Faculty of Computing and Data Science
Sep 2023 - Jun 2027 (Expected)
- Relevant coursework: Supervised/Unsupervised Learning, Neural Networks, Data Mining, Search & Logic, Probability & Statistics, Linear Algebra, SQL, Cloud Computing

EXPERIENCE

Training Program - Advanced DS & AI - TechTrek | Alexandria University **Aug 2026 – Sep 2026**
- Completed intensive training in machine learning, deep learning, NLP, SQL, cloud computing, deployment, and MLOps, covering CNNs, LSTMs, and Transformers.
- Applied model evaluation, cross-validation, hyperparameter tuning, and deployment through practical projects and exercises.

Field Training - IoT Systems Development | Alexandria University **Jul 2025 - Aug 2025**
- Developed IoT systems using **ESP32, sensors, PlatformIO, and Wokwi**, applying MQTT, I2C, SPI, and UART with HiveMQ and Supabase integration.
- **Led a team capstone** building a Smart Pet Care System using ESP32 (C), Flutter, and Flask for real-time monitoring and alerts.

PROJECTS

Intelligent Traffic Sign Recognition System | Capstone Project - AI Training Team Project **Sep 2026**
- Built an end-to-end 43-class GTSRB recognition system, comparing classical ML, a custom CNN, and MobileNetV3-Small transfer learning.
- Achieved 96.97% Top-1 accuracy and 95.78% Macro F1 with the custom CNN, evaluating Top-5 accuracy, robustness, latency, and throughput.
- Deployed optimized TensorFlow Lite Float16/Int8 models (475 KB/254 KB) with a Streamlit interface featuring Top-3 predictions and Grad-CAM explanations.

Medical Agentic RAG | Personal Project **Sep 2026**
- Built an agentic RAG Telegram assistant for Arabic/English medical information, combining LLM-based query routing, evidence-grounded retrieval, and clinic service workflows.
- Implemented Qdrant retrieval with multilingual embeddings, query rewriting, relevance gating, clarification loops, and input/output guardrails using LiteLLM, Groq, and Hugging Face.
- Deployed the system with FastAPI, Telegram webhooks, and Vercel, with persistent sessions and booking state managed through Qdrant.

Egypt & KSA Job Market Analysis | Team Project **May 2025**
- Collected and processed 9,500+ job listings using Python web scraping and an automated SQL pipeline, uncovering seasonal hiring trends and high-growth sectors in Egypt and Saudi Arabia.
- Built 12 visualizations with Plotly & Seaborn to present market dynamics and employer insights.

SKILLS

Programming: Python, SQL, Java, R
ML/DL: scikit-learn, TensorFlow, PyTorch, Keras, CNNs, LSTM, Transformers, Transfer Learning
Computer Vision: OpenCV, Image Classification, Grad-CAM, Image Preprocessing
Data: Pandas, NumPy, EDA, Statistics, Data Visualization
Deployment: Streamlit, Flask, TensorFlow Lite, AWS
Tools: Git, Linux, Jupyter, PlatformIO, Wokwi
Other: Web Scraping, IoT, MQTT, Flutter

CERTIFICATIONS

Machine Learning Specialization – Stanford & Coursera | AWS Academy Cloud Foundations